

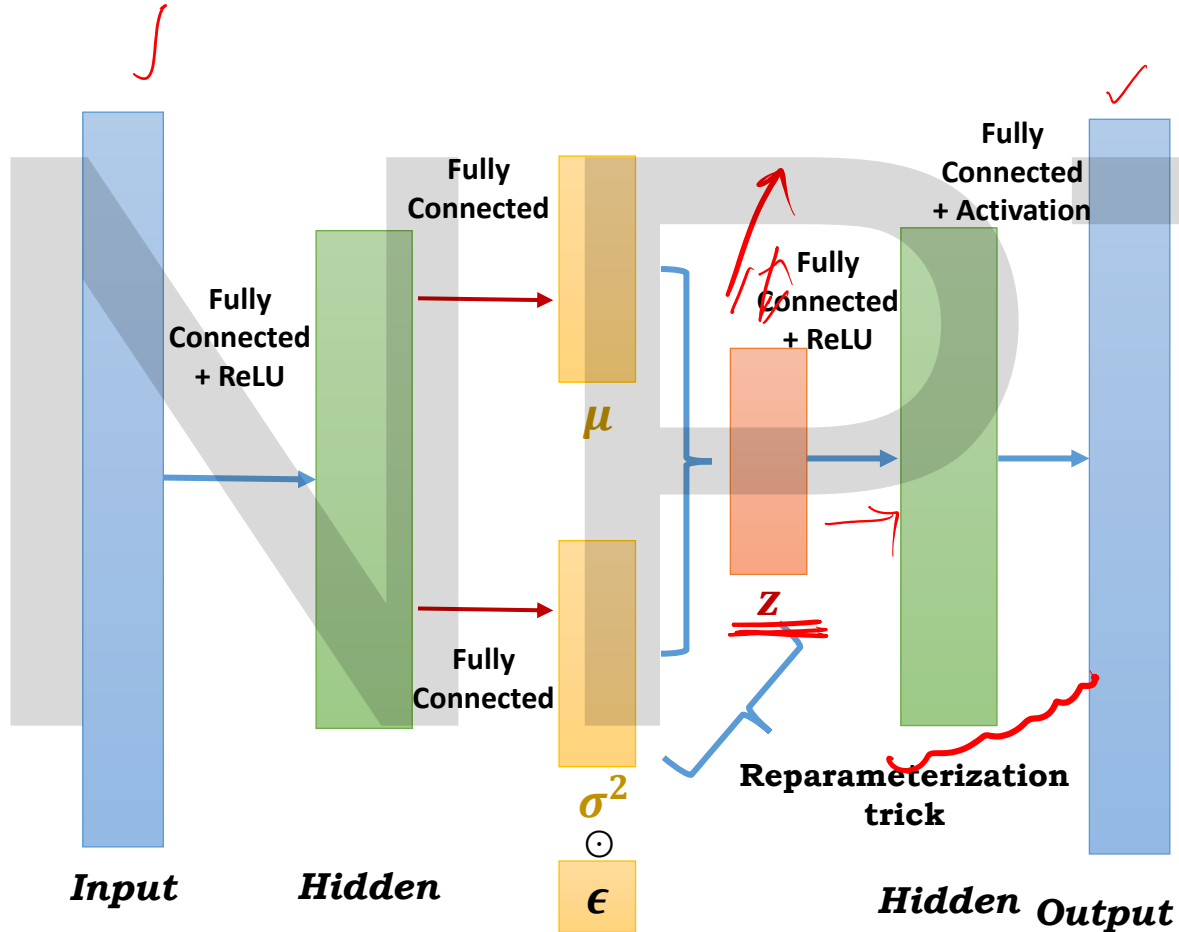
Course: Fundamentals of Generative AI and Large Language Models: Theory and Practice

Week 4: Advanced Variational Autoencoders & Practical Applications

Today's Session Overview

- Recap of VAE
- Conditional VAE Architecture
- cVAE Loss Function
- Numerical Example

VAE recap



So, the standard VAE learns:

$$x \rightarrow z \rightarrow \hat{x}$$

In a standard VAE, the model is trained only with the input data x .

The VAE learns a compressed latent representation z and reconstructs the input as $\hat{x}$.

The **encoder** converts input data into a probabilistic latent representation. It maps input x (like an image) to a distribution (mean μ , variance σ^2) over the latent space.

Once the encoder gives $\mu_\phi(x)$ and $\sigma_\phi^2(x)$ of the approximate posterior distribution,

$$q_\phi(z | x) = \mathcal{N}(\mu_\phi(x), \sigma_\phi^2(x))$$

The **latent vector** z is sampled using the reparameterization trick:

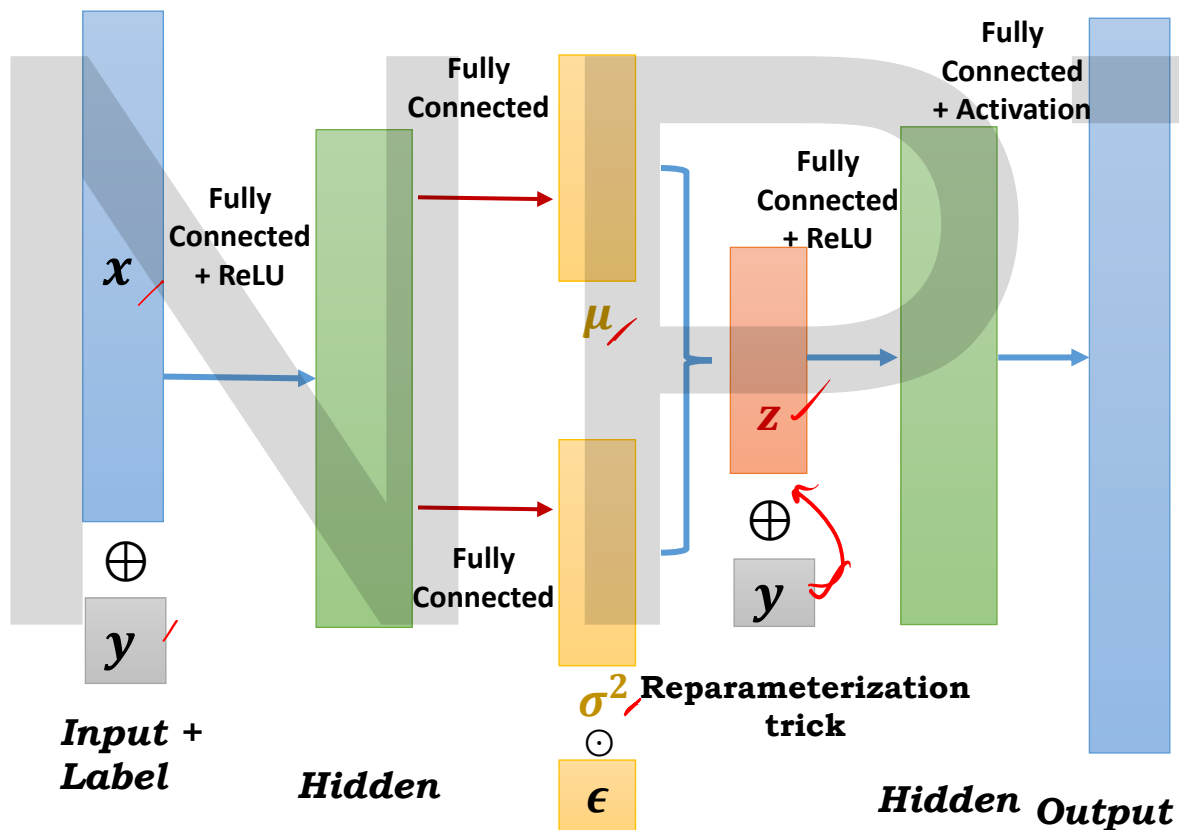
$$z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon \quad \epsilon \sim \mathcal{N}(0, I)$$

The **decoder** takes a latent variable z and reconstructs or generates data. It maps the latent vector, a compressed version of the data, back to the original data space. The function is denoted as:

$$p_\theta(x | z)$$

where x is the generated data.

Conditional VAE (cVAE)



$$z = \mu_{\phi}(x, y) + \sigma_{\phi}(x, y) \odot \epsilon$$

In a standard VAE, generation is not explicitly controlled.

The model learns:

$$x \rightarrow z \rightarrow \hat{x}$$

But if we want to generate a specific type of output, we need extra guidance.

For example:

Generate digit 5

or

Generate a cat image

A **Conditional VAE** is an extension of VAE where we give an extra input called a **condition** y , such as a class label or attribute.

In a **cVAE**, the model learns:

$$(x, y) \rightarrow z \rightarrow \hat{x}$$

Loss Function in cVAE

$$L(x) = -\mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] + D_{KL}(q_{\phi}(z|x) \parallel p(z))$$

$$L_{cVAE}(x) = -\mathbb{E}_{q_{\phi}(z|x,y)}[\log p_{\theta}(x|z,y)] + D_{KL}(q_{\phi}(z|x,y) \parallel p(z))$$

std VAE
Loss funⁿ

Numerical Example

$$L(x) = -\mathbb{E}_{q_{\phi}(z | x)}[\log p_{\theta}(x | z)] + D_{KL}(q_{\phi}(z | x) \parallel p(z))$$

$$L_{cVAE}(x) = -\mathbb{E}_{q_{\phi}(z | x, y)}[\log p_{\theta}(x | z, y)] + D_{KL}(q_{\phi}(z | x, y) \parallel p(z))$$

In VAE, encoder input is only:

$$x = [5.1, 3.5, 1.4, 0.2]$$

In cVAE, encoder input becomes:

$$[x, y]$$

Assume the class label is:

$$y = [1, 0]$$

So the conditional encoder input is:

$$[x, y] = [5.1, 3.5, 1.4, 0.2, \underline{1}, \underline{0}]$$

How label changed the hidden representation

In the original VAE, the hidden activation was:

$$a_{VAE} = [1.07, 1.93, 0.87]$$

In cVAE, after adding label $y = [1, 0]$:

$$a_{cVAE} = [1.27, 1.83, 1.17]$$

This is the key idea:

The label contributes additional weighted information to the encoder hidden layer

Numerical Example

$$L(x) = -\mathbb{E}_{q_{\phi}(z | x)}[\log p_{\theta}(x | z)] + D_{KL}(q_{\phi}(z | x) \parallel p(z))$$

$$L_{cVAE}(x) = -\mathbb{E}_{q_{\phi}(z | x, y)}[\log p_{\theta}(x | \underline{z}, y)] + D_{KL}(q_{\phi}(z | x, y) \parallel p(z))$$

In cVAE, decoder receives both z and y .
So decoder input is:

$$z = [0.917, 0.153]$$

$$[z, y] = [0.917, 0.153, 1, 0]$$

Input to the Decoder: [0.917, 0.153, 1, 0]

Decoder
z → *y* → $\hat{x}$

This tells the decoder:

**generate/reconstruct x using latent information z ,
but conditioned on class label $y = [1, 0]$.**

Numerical Example

$$L(x) = -\mathbb{E}_{q_{\phi}(z | x)}[\log p_{\theta}(x | z)] + D_{KL}(q_{\phi}(z | x) \parallel p(z))$$

$$L_{cVAE}(x) = -\mathbb{E}_{q_{\phi}(z | x, y)}[\log p_{\theta}(x | z, y)] + D_{KL}(q_{\phi}(z | x, y) \parallel p(z))$$

VAE reconstructed output:

$$\hat{x}_{VAE} = [0.103, -0.028, 0.206, 0.243]$$

cVAE reconstructed output:

$$\hat{x}_{cVAE} = [0.147, -0.047, 0.280, 0.337]$$

This shows that the label y changes the decoder hidden representation and therefore changes the reconstructed output.

Next Session

➤ Latent space interpolation

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Thank you